EXPERIMENT NO. 5

NAME OF EXPERIMENT: Study of JFET Characteristics Using PSpice

AIM: To study the input and output characteristics of an N-channel Junction Field Effect Transistor (JFET) using PSpice.

SOFTWARE REQUIRED: PSpice Simulator.

COMPONENTS REQUIRED:

- N-channel JFET (e.g., J201 or J2N3819 model available in PSpice)
- DC Voltage Sources
- Resistors (1K)
- Ground

THEORY:

A Junction Field Effect Transistor (JFET) is a voltage-controlled semiconductor device in which the drain current is controlled by the gate-to-source voltage (VGS). It is a unipolar device because current conduction occurs due to majority charge carriers only.

The two important characteristics of a JFET are:

1. Drain Characteristics

It shows the variation of Drain Current (ID) with Drain-to-Source Voltage (VDS) for different values of Gate-to-Source Voltage (VGS).

As VDS increases, ID increases initially and then becomes nearly constant after reaching the pinch-off region.

2. Transfer Characteristics

It shows the variation of Drain Current (ID) with Gate-to-Source Voltage (VGS) while keeping VDS constant.

The transfer characteristic follows Shockley's equation:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Where:

ID = Drain Current
IDSS = Drain saturation current at VGS = 0 V
VP = Pinch-off Voltage

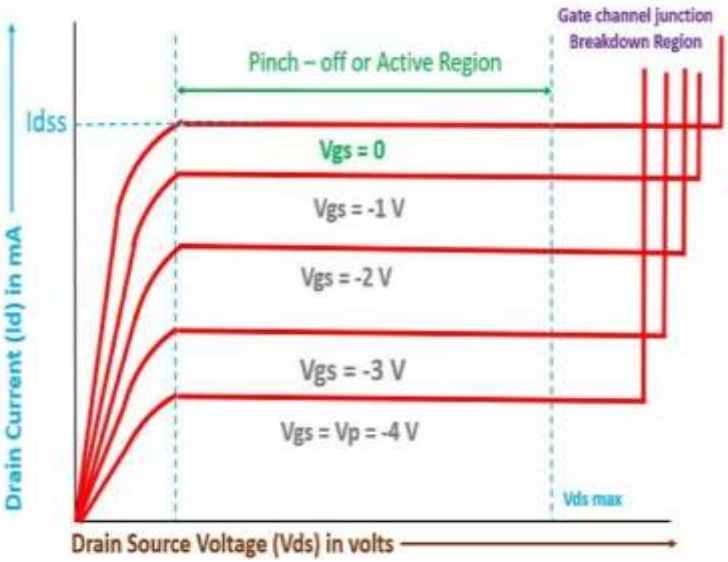


Fig: Drain Characteristics of JFET

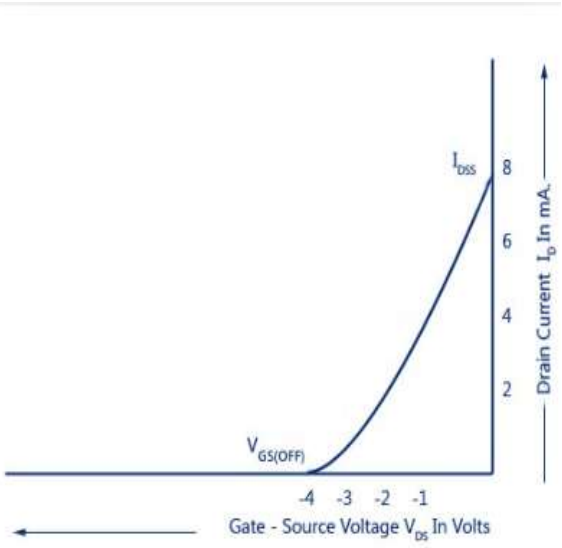


Fig: Transfer Characteristics of JFET

CIRCUIT DIAGRAM:

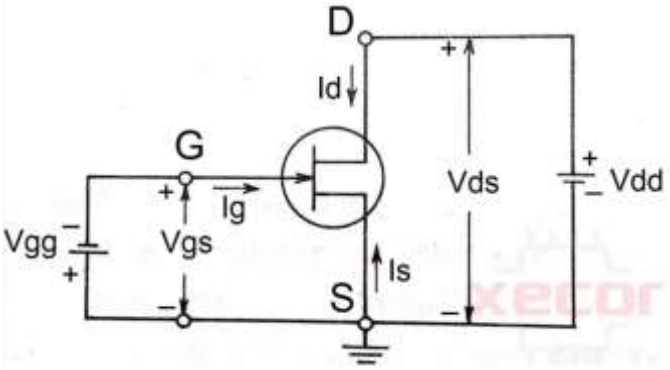


Fig: N-Channel JFET

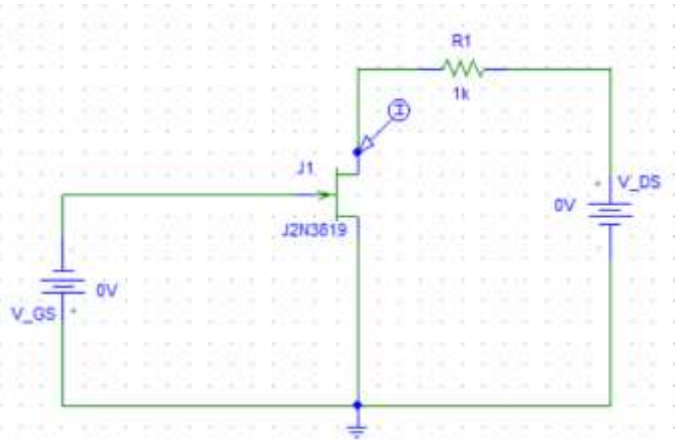


Fig: N-channel JFET with Pspice

PROCEDURE:

A. Drain Characteristic

1. Open PSpice and create a new project.
2. Place the N-channel JFET from the component library and connect component as per circuit diagram.

3. Use one voltage source for (V_DS) and another voltage source for (V_GS).
4. Verify the polarity of the voltage sources, particularly for the negative (V_GS) values.
5. Set the Gate-to-Source Voltage values as V_GS=0V; -1V; -2V; -3V.
6. Open DC Sweep Analysis
    - Select: **Analysis → Setup → DC Sweep**
    - Enable **DC Sweep**.
7. Set the (V_DS) Sweep
    - Swept Var Type: Voltage Source
    - Sweep Type: Linear
    - Name: V_DS
    - Set:
        - Start Value: 0 V
        - End Value: 40 V
        - Increment: 5 V
8. Set Nested Sweep
    - Select **Nested Sweep** in the DC Sweep setup.
    - Swept Var Type: Voltage Source
    - Sweep Type: Linear
    - Name: V_GS

    Enter the required:
    - Start value: 0
    - End value: 4
    - Increment/Step value:1
    - Enable Nested Sweep

    Click OK and close the setup window.
9. Run the Simulation
10. Observe the ohmic/linear region and pinch-off region of the JFET from the characteristics.

B. Transfer Characteristics

1. Open the same JFET circuit in PSpice.
2. Check all circuit connections and voltage-source values.
3. Keep (V_DS) Constant
    - Set the drain-to-source voltage to a constant value. i.e. V_DS=10v

4. Open DC Sweep Analysis

    - Select: **Analysis → Setup → DC Sweep**
    - Enable **DC Sweep**.

5. Set the (V_GS) Sweep

    - Select the (V_GS) voltage source as the sweep variable.
    - Swept Var Type: Voltage Source
    - Sweep Type: Linear
    - Name: V_GS
    - Set:
        - Start Value: 0 V
        - End Value: 5 V
        - Increment: 0.1 V.

6. Run the Simulation

    - Click **OK** and close the setup window.
    - Click **Simulate**.

7. Plot the Transfer Characteristics

    - Plot the drain current (I_D) against gate-to-source voltage (V_GS):

8. Determine (I_DSS)

    From the graph, observe the value of drain current at V_GS=0V i.e. I_DSS.

9. Determine Pinch-Off Voltage (Vp or V_GS(off))

    From the transfer characteristic, identify the value of (V_GS) at which the drain current approaches zero.

OBSERVATION TABLE:

A. Drain Characteristics

VGS(V)	VDS(V)	ID(mA)
0	0-40	
-1	0-40	
-2	0-40	
-3	0-40	

B. Transfer Characteristics: V_DS=10V

V_GS(V)	ID(mA)
0	
-1	
-2	
-3	

C. IDSS= _____ mA

D. Pinch_off voltage (V_GS (off) or Vp) = _____ V

CALCULATION:

The Drain current of JFET is given as

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

IDSS= ——— mA (from graph)

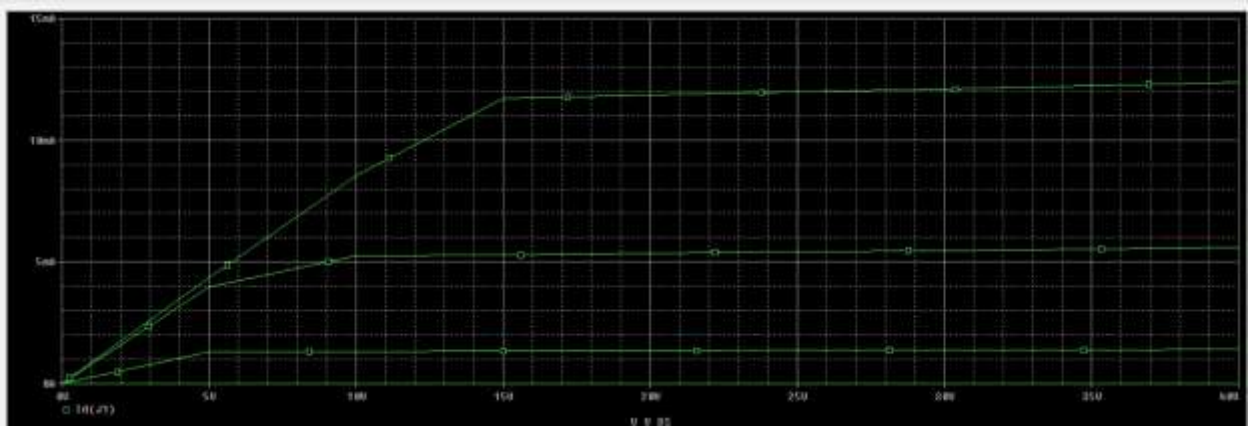
Vp= ——— V (from graph)

Calculate Drain current (I_D) for different values of V_{GS}

V_{GS}(V)	I_D(mA) (Calculated)
0	
-1	
-2	
-3	

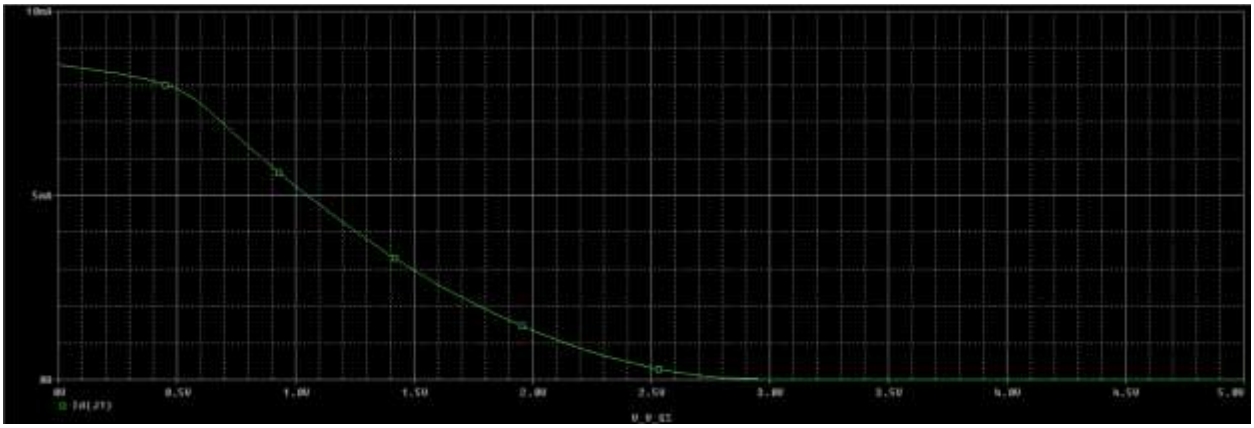
OUTPUT:

A. Drain Characteristics of JFET



Drain Characteristics: ID versus VDS for different values of VGS.

B. Transfer characteristics of JFET:



Transfer Characteristics: ID versus VGS at a constant VDS

RESULT

The drain and transfer characteristics of the N-channel JFET were successfully obtained using PSpice. The pinch-off region, saturation current (IDSS), and pinch-off voltage (VP) were observed from the simulated characteristics of the JFET.

CONCLUSION

VIVA VOCE:

1. What is a junction field effect transistor or JFET?
2. Describe the working of junction field effect transistors.
3. What are the three main operating regions of JFET?
4. What is the difference between a BJT and a JFET?
5. What are the advantages of JFET?